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¹) Baker, Morya. "1,500 scientists lift the lid on reproducibility." Nature, no. 533 (May 26, 2016): 452-54. doi:10.1038/533452a.

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ARTICLE

Pilot-scale demonstration of an end-to-end integrated and continuous biomanufacturing process

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Abstract

There has been increasing momentum recently in the biopharmaceutical industry to transition from traditional batch processes to next-generation integrated and continuous biomanufacturing. This transition from batch to continuous is expected to offer several advantages which, taken together, could significantly improve access to biologics drugs for patients. Despite this recent momentum, there has not been a commercial implementation of a continuous bioprocess reported in the literature. In this study, we describe a successful pilot-scale proof-of-concept demonstration of an end-to-end integrated and continuous bioprocess for the production of a monoclonal antibody (mAb). This process incorporated all of the key unit operations found in a typical mAb production process, including the final steps of virus removal filtration, ultrafiltration, diafiltration, and formulation. The end-to-end integrated process was operated for a total of 25 days and produced a total of 4.9 kg (200 g/day or 2 g/L BRX/day) of the drug substance from a 100-L perfusion bioreactor (BRX) with acceptable product quality and minimal operator intervention. This successful proof-of-concept demonstrates that end-to-end integrated continuous bioprocessing is achievable with current technologies and represents an important step toward the realization of a commercial integrated and continuous bioprocessing process.

KEYWORDS

continuous bioprocessing, continuous UF/DF, continuous virus filtration, end-to-end, integrated continuous biomanufacturing

1 | INTRODUCTION

The current standard for the commercial manufacturing of biologics drugs is the use of batch processes in large, centralized stainless-steel facilities. Similar to other industries, there has been a push to transition biologics manufacturing away from batch processing toward integrated and continuous processing (Badman et al., 2019). The transition from traditional batch processing to integrated and continuous biomanufacturing is expected to yield a number of benefits, including decreased

capital and operating costs; improved productivity and decreased facility size through process intensification; enabling production in multiproduct “ballroom” facilities; improved flexibility to respond to changing demand; and improved consistency of product quality (Fisher et al., 2019; Konstantinov & Cooney, 2015; Pollock et al., 2017; Walther et al., 2015, 2019). The net effect of these benefits would be the potential to drastically improve access to life-saving biologics medicines, including in developing countries where patients are currently distressingly underserved (Rathore & Shereef, 2019; Saqib et al., 2018).

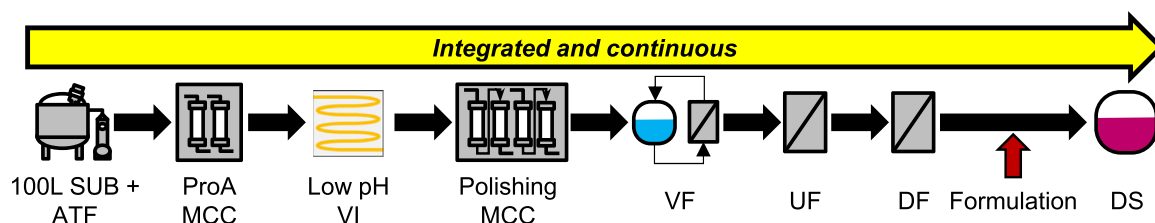


FIGURE 1 Schematic of the end-to-end continuous process in this demonstration. Each unit operation was physically integrated and product flowed continuously from one unit to the next. AFT, alternating tangential flow filter; DF, diafiltration; DS, drug substance; MCC, multi-column chromatography; ProA, Protein A; UF, ultrafiltration; VF, virus filtration; VI, virus inactivation

Several significant advances in enabling technologies for integrated and continuous bioprocessing (ICB) have been reported in the last few years (Farid, 2019; Godawat et al., 2012; Jungbauer, 2019; Mothes et al., 2016; Orozco et al., 2017; Pollock et al., 2013; Rucker-Pezzini et al., 2018). In particular, the integration of a production bioreactor with one or more continuous chromatography steps and low pH virus inactivation has reached a stage of technical maturity that several companies are moving to clinical and commercial manufacturing with these hybrid ICB systems (Brower et al., 2019; David et al., 2020; Hewig, 2019; Orozco et al., 2019; Otes et al., 2020; Shields, 2019; Shultz, 2019). This transition has been supported by several economic analyses that demonstrated the value of a hybrid continuous process (Pollock et al., 2017; Walther et al., 2015). However, integration of the virus removal filtration and final ultrafiltration/diafiltration operations is less well reported in the literature. There may be several reasons for this gap, including concerns around the long-term operation of virus removal filtration and a lack of suitable technology for single-pass diafiltration (SP-DF). Incorporation of these unit operations would enable a truly end-to-end drug substance (DS) production system, with the potential opportunity to further reduce manufacturing facility size and cost.

Building on previous demonstrations (Godawat et al., 2015; Warikoo et al., 2012), in this study, we describe a proof-of-concept for a pilot-scale, end-to-end integrated and continuous DS production system, using a mAb as a model protein. This system integrates all unit operations for a typical mAb DS process. As shown in Figure 1, this integrated process consisted of a perfusion bioreactor, multi-column capture and polishing chromatography, low pH virus inactivation, virus removal filtration, ultrafiltration, diafiltration, formulation (excipient addition), and sterile filtration. Several key design goals, focused on automation, process closure, continuous operation, and process scale, were incorporated into the process design to ensure that this proof-of-concept would inform the design and implementation of a future good manufacturing practice (GMP) system. These design goals are highlighted in Table 1 and were applied both to individual unit operations as well as the system overall. The results of the proof-of-concept described here signify a significant step forward in the development and implementation of integrated and continuous bioprocessing.

TABLE 1 Design goals for the end-to-end continuous process demonstration

Category	Design goals
Automated	- Minimize manual interventions - Ensure process consistency
Closed	- Enable operation in a “ballroom” CNC space - Minimize in-suite preparation steps
Continuous	- Match volume and mass flux between integrated operations - Minimize product residence time - Decrease or eliminate surge vessels
Process Scale	- Identify relevant scale-dependent failure modes - Test at-scale automation code and process closure

Abbreviation: CNC, controlled non-classified.

2 | MATERIALS AND METHODS

2.1 | Continuous process system

The end-to-end process for this study consisted of the integration of a perfusion bioreactor (BRX), multi-column chromatography (MCC), in-line virus inactivation (VI), virus filtration (VF), ultrafiltration (UF) and diafiltration (DF), in-line formulation, and final filtration to generate DS.

2.2 | Integration and closure

All unit operations were physically integrated and flows were controlled to ensure no accumulation of mass or volume in the system. As an added precaution during the proof-of-concept, several surge vessels were placed throughout the system to smooth any temporary perturbations in flow and to allow time to react to process disruptions. Surge vessels were present between the following operations: the bioreactor and capture chromatography, capture chromatography and virus inactivation, virus inactivation and polishing chromatography, polishing chromatography and virus filtration, and virus filtration and ultrafiltration.

To ensure process closure, all product contact materials were either sterilized (via gamma irradiation or autoclave), steamed-in-place, or sanitized before installation. The Protein A (ProA) columns were sterilized via gamma irradiation, as described previously (Varner et al., 2020). The use of sanitized components was minimized where possible. Details of how each individual component was sterilized or sanitized can be found in the Supporting Information. After sterilization or sanitization of individual components, all connections were made via aseptic welding.

2.3 | Cell culture

The target molecule for this study was generated via a Chinese Hamster Ovary (CHO) cell culture producing a recombinant IgG in a 100-L perfusion bioreactor (BRX). The perfusion bioreactor was integrated with an alternating tangential flow (ATF) system for cell removal, as previously described (Warikoo et al., 2012). The cell-free, product-containing harvest was collected in an integrated surge vessel before capture chromatography.

2.4 | Chromatography

A product containing harvest was continuously captured on 2×1 -L MabSelect Sure (MSS) Protein A (ProA) Ready to Process columns (Cytiva). The two columns were operated in parallel in bind-and-elute mode, with harvest continuously flowing to one of the two columns. Column switching was based on a 3% breakthrough of Δ UV signal with no shared loading (Godawat et al., 2012).

After virus inactivation, polishing chromatography was performed using Capto Adhere and Capto Phenyl resin (Cytiva) in 0.2 L (5×20 cm) Opus prepacked columns (Repligen), operated in flow-through mode. The Capto Adhere flowthrough was loaded directly onto the Capto Phenyl column, with no interstep adjustment. Two Adhere/Phenyl trains were operated in parallel, with virus inactivated material continuously flowing to one of the two trains. A schematic of the polishing column setup can be found in Figure 2. Column switching was based on the cadence of the ProA operation, with two ProA cycles coinciding with one Adhere/Phenyl cycle.

All chromatography operations were performed on a pilot-scale, steam-in-place (SIP), four-column periodic counter-current chromatography (PCC) skid (Cytiva). The MSS columns occupied two of the four-column positions, while the other two positions were occupied by the Adhere/Phenyl column pairs.

2.5 | Virus inactivation (VI)

The capture column eluate was homogenized in a 3-L mixing vessel (Eppendorf) before virus inactivation. The homogenized capture column eluate was continuously removed from the mixing vessel with a peristaltic pump (Watson Marlow) and acidified by an in-line

addition of acetic acid to a target pH of 3.6. Mixing for this and all subsequent mixing steps was via a helical static inline mixer (Koflo). The acidified stream was pumped through a custom-made tubular flow reactor (TFR) designed to target a minimum residence time (MRT) of 30 min. After passing through the TFR, the virus inactivated product stream was adjusted by an in-line addition of sodium acetate for subsequent processing. Acid and base titrations were performed based on a feed-forward calculation of the incoming process stream. The pH was monitored but not directly controlled.

2.6 | Virus filtration (VF)

The Capto Phenyl flowthrough was processed through a $0.5/0.1 \mu\text{m}$ SHR filter (Millipore Sigma) before processing through a 0.1 m^2 Planova 20 N virus removal filter (Asahi Kasei Bioprocess). The virus removal filter (VRF) was operated in tangential flow mode, with the VRF located in a pressurized loop consisting of a vessel, a recirculation pump, and a $0.5/0.2 \mu\text{m}$ SHC filter (Millipore Sigma). The flux through the VRF was controlled by adjusting the pressure in the loop with compressed air. A schematic of the VRF setup can be found in Figure 3.

2.7 | Ultrafiltration and diafiltration (UF/DF)

The output from the VRF was concentrated with a 0.065 m^2 30 kDa cutoff Cadence Inline Concentrator (Pall) UF cassette. The outlet product concentration from the UF stage was controlled with a PID feedback loop via an inline concentration measurement using a FlowVPE (C Technologies). The outlet from the UF cassette was subsequently buffer exchanged into an acetate buffer with a prototype SP-DF cassette (30 kDa cutoff) from Sartorius. The target DF ratio was maintained by controlling the relative flow rates of the diafiltration buffer and product.

2.8 | Formulation and final filtration

Following buffer exchange, the product was formulated in-line with a concentrated excipient stock solution containing sugar and polysorbate 80 to generate formulated bulk. The formulated bulk was pumped through a sterilizing grade $0.5/0.2 \mu\text{m}$ SHC filter (Millipore Sigma) to generate a DS. Each batch of DS was collected in a separate 20- or 50-L bioprocess container (Thermo Fisher Scientific).

2.9 | Buffer preparation

The process solutions for the VI, polishing chromatography, and UF/DF operations were generated from single-component stocks using an in-line conditioning skid (Asahi Kasei Bioprocess). The process solutions for all other unit operations were manually prepared.

2.10 | Analytical methods

In-house assays were used for the quantitation of protein concentration, host cell proteins (HCPs), high molecular weight species (HMWS), residual Protein A (rProA), purity, and charge isoform distribution. Product concentration was measured via absorbance at 280 nm using a SoloVPE (C Technologies). rProtA and HCPs were quantitated by ELISA using commercial kits (Cygnus Technologies). HMWS levels were measured by HPLC-SEC using a TSK-Gel G3000SW_{XL} column (Tosoh Bioscience). Purity was measured using an in-house sodium dodecyl sulfate-capillary electrophoresis (CE-SDS) method under nonreducing (NR) conditions. Charge isoform distribution was determined by capillary isoelectric focusing (cIEF) using an in-house method. Compendial methods were used for the determination of pH (USP <791>), osmolality (USP <785>), and bioburden (USP <51>).

2.11 | Process monitoring

Process flows were monitored using single-use ultrasonic flowmeters (Levitronix). The pressure was monitored with single-use pressure sensors (Pendotech). The pH of the VI stream was monitored using in-line pH meters (Hamilton). The protein concentration of the VI stream was monitored using an on-line UV absorbance sensor and photometer (Pendotech). The pH, conductivity, and UV absorbance of the chromatography operations were monitored with the respective built-in sensors on the PCC skid. The protein concentration in the UF stream was monitored via absorbance at 280 nm using a FlowVPE (C Technologies).

2.12 | Control and data management

All chromatography unit operations were controlled using UNICORN software (Cytiva). Control of the nonchromatography unit operations (VI, VRF, UF/DF, and formulation) was achieved with a DeltaV Distributed Control System (Emerson). All in-line sensor and process data were collected and stored in a PI data historian (OSIsoft).

3 | RESULTS AND DISCUSSION

3.1 | Process integration

The primary objective of the current study was to demonstrate the feasibility of manufacturing a therapeutic antibody at the pilot-scale with a fully integrated and continuous process. To achieve this, this demonstration incorporated all the key unit operations included in a typical mAb production process, including a production bioreactor, capture chromatography, virus inactivation, polishing chromatography, virus removal filtration, and ultrafiltration/diafiltration. A schematic of the overall production process can be found in Figure 1.

Each of the unit operations was physically integrated, and process flows were controlled throughout the system to ensure no accumulation of the material.

The end-to-end integrated process, from the bioreactor to DS, was operated for a total of 25 days. For the proof-of-concept, individual unit operations were brought online sequentially and allowed to reach steady-state before initiating the next operation. In future runs, all the unit operations would be initiated at the start of the campaign. The startup and integration timing are summarized in Figure 4a. After startup, the BRX, ProA, VI, and polishing operations were operated until the end of the campaign. The VF, UF/DF, and DS operations were terminated early, as described in subsequent sections.

The average product residence time for each downstream unit operation is shown in Figure 4b. The solid shaded region is the average residence time in the operation itself, while the check shaded region indicates the average residence time in the surge vessel preceding the operation. The average residence time in the surge vessel between the bioreactor and capture is not included. Overall, the average residence time from capture to DS was approximately 12 h. For this setup, the theoretical minimum residence time (with no surge vessels) would be approximately 7 h.

3.2 | Process closure and bioburden control

One of the goals of this demonstration was to demonstrate that an end-to-end integrated continuous process could be operated in a closed manner such that the process could be implemented in a controlled nonclassified (CNC) ballroom facility. To achieve this, each individual unit operation was designed to enable closed processing, rendered bioburden free through sterilization or sanitization, and connected aseptically to the integrated system, as described in Section 2 and the Supporting Information.

The overall bioburden control strategy for this run was based on the premise that the risk of bioburden ingress was rendered negligible by virtue of the system being operated in a closed manner. Because of this, no intermediate flushing of system components in caustic was performed. To confirm that the system remained bioburden free, samples were taken on Day 42 of the campaign at five points in the process and analyzed for bioburden. The sampling points were post-ProA, post-VI, post-polishing, post-VF, and post-excipient addition. For all samples other than the post-excipient addition sample, the bioburden value was <1 CFU/10 ml. For the post-excipient addition sample, there was a significant presence of yeast colonies (~2000 CFU/10 ml). No definitive root cause was determined for the positive bioburden test. However, it is suspected that a sampling error was the likely root cause, as the operator noted a loss of integrity of the sample bag during the sampling process. Although no subsequent samples were submitted for bioburden testing, there was no indirect evidence of bioburden ingress (e.g., pressure buildup on the final sterilizing grade filter) over the course of ~20 additional days of processing.

3.3 | Detailed performance for individual unit operations

Key performance details for individual unit operations can be found in the following sections. The product quality of the process intermediates is presented in Section 3.4.

3.3.1 | ProA capture

The ProA capture operation was integrated with the bioreactor on Day 12 and executed for a total of 58 days. Over the course of the 58 days, each of the two columns was cycled ~190 times, with an average cycle time of approximately 3.5 h. Chromatograms and elution peak overlays for the ProA operation can be found in Figure 2. For clarity, only a portion of the campaign (corresponding to the first 7 days of DS generation) is shown in the chromatogram, and every five elution cycles are included in the overlay.

For the first ~154 cycles, the elution peak overlays were extremely consistent, with only a small number (<5) of outlier peaks. A known process disturbance caused a shift in peak shape around Cycle 155 (~Day 60). The details are captured in the process disturbance section (Section 3.6.2).

The columns were consistently loaded to 3% breakthrough of the delta UV signal, which corresponded to an average binding capacity of approximately 48 g/L resin. There was a slight decrease in binding capacity due to cycling, from approximately 53 to 46 g/L over

the first 175 cycles, which is consistent with the expected decline based on the manufacturer's data. Overall, the ProA performance was consistent and within expectations for the entire campaign.

3.3.2 | Low pH virus inactivation

For this demonstration, an in-line virus inactivation with a defined residence time module was selected over other alternatives (including an automated two-tank VI) as it was a better fit with the overall design goals. A schematic representation of the VI operation is found in Figure 5a. Before integration, off-line testing of the residence time module was conducted to confirm that the minimum residence time was >30 min at the expected flow rates for the integrated operation (data in the Supporting Information).

The low pH VI operation was integrated with the ProA operation on Day 15. During the first week, the in-line acid and base addition control strategies were adjusted to ensure robust operation. The virus inactivation pH reached a steady state on Day 23 and continued until the end of the campaign (Day 70). The pH was monitored at two points in the virus inactivation stream: after acid addition and after base addition. The pH trends for the virus inactivation operation can be found in Figure 5.

Overall, the inactivation pH was well-controlled within the target ranges for the duration of the campaign. The pH deviated above the maximum tolerance for inactivation (pH 3.85) at two

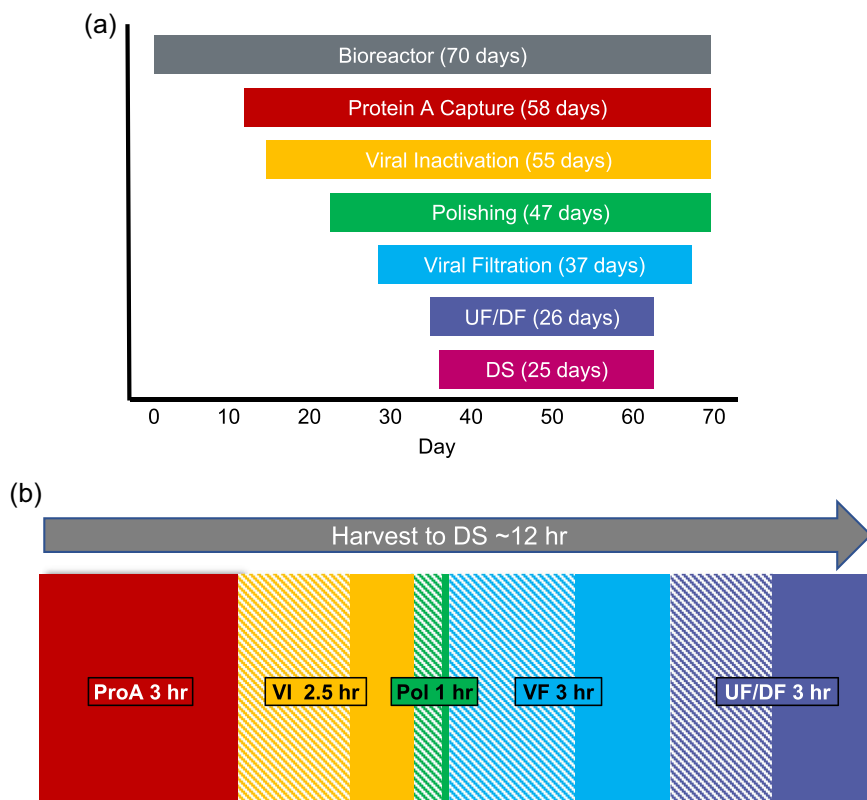


FIGURE 2 Summary of process integration. (a) Startup timing and duration for individual unit operations. (b) Average product residence time in each unit operation. The residence time in the actual operation is shown in the solid color, the residence time in the surge vessel preceding the operation is shown in the hatched color. DF, diafiltration; DS, drug substance; ProA, Protein A; UF, ultrafiltration; VI, virus inactivation

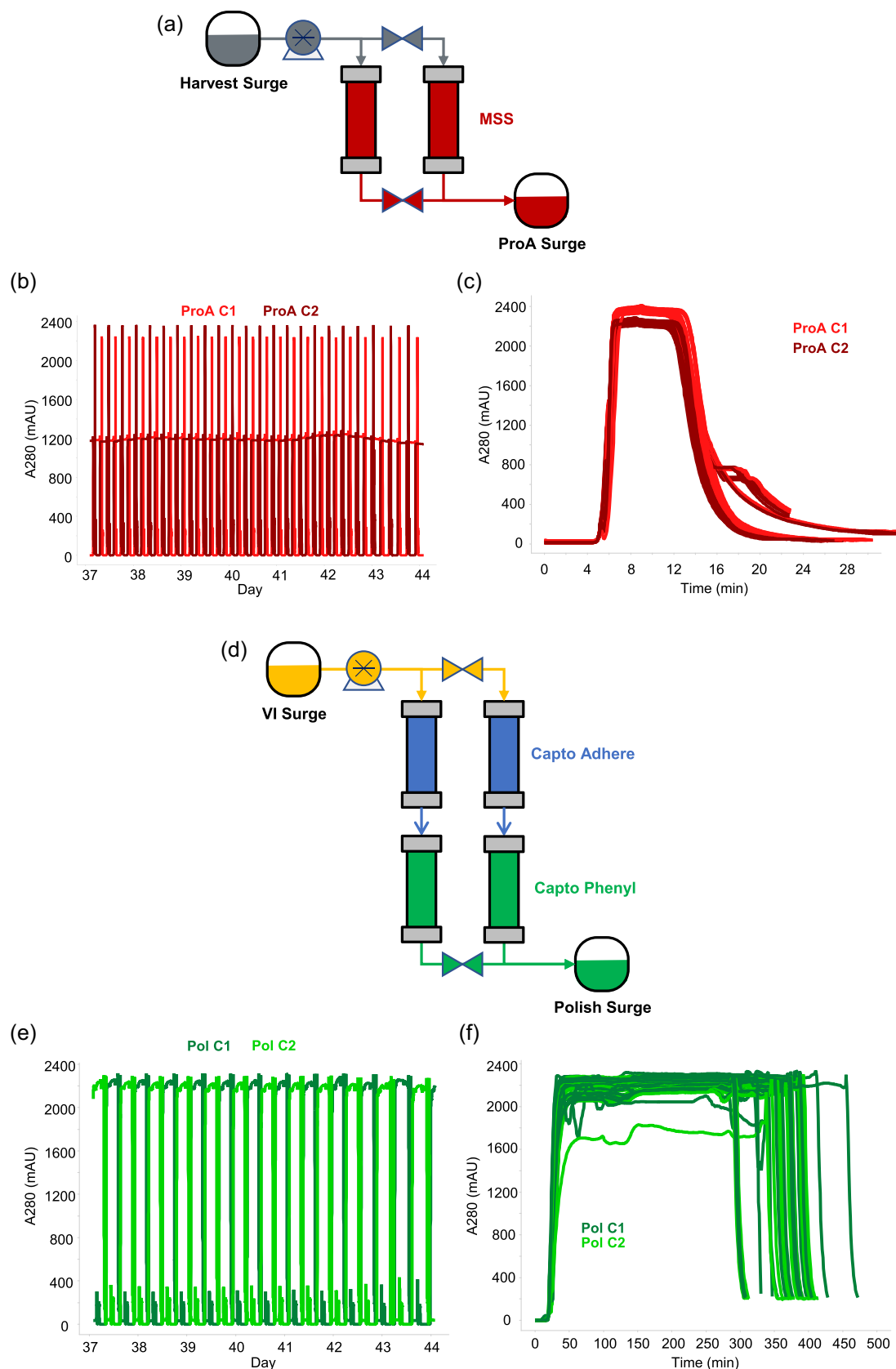


FIGURE 3 Summary of results for the chromatography operations. (a) Schematic of the multi-column ProA operation. (b) Snapshot of UV chromatogram for 7 days of operation for ProA. (c) Chromatogram overlay of every five ProA elution peaks. The process disturbance in the elution buffer is evidenced by the change in the shape of the elution peak tail. (d) Schematic of the multi-column polishing operation. (e) Snapshot of UV chromatogram for 7 days of operation for polishing columns. (f) Polishing column flowthrough peak overlay showing every five cycles. ProA, Protein A; VI, virus inactivation; Pol, polishing

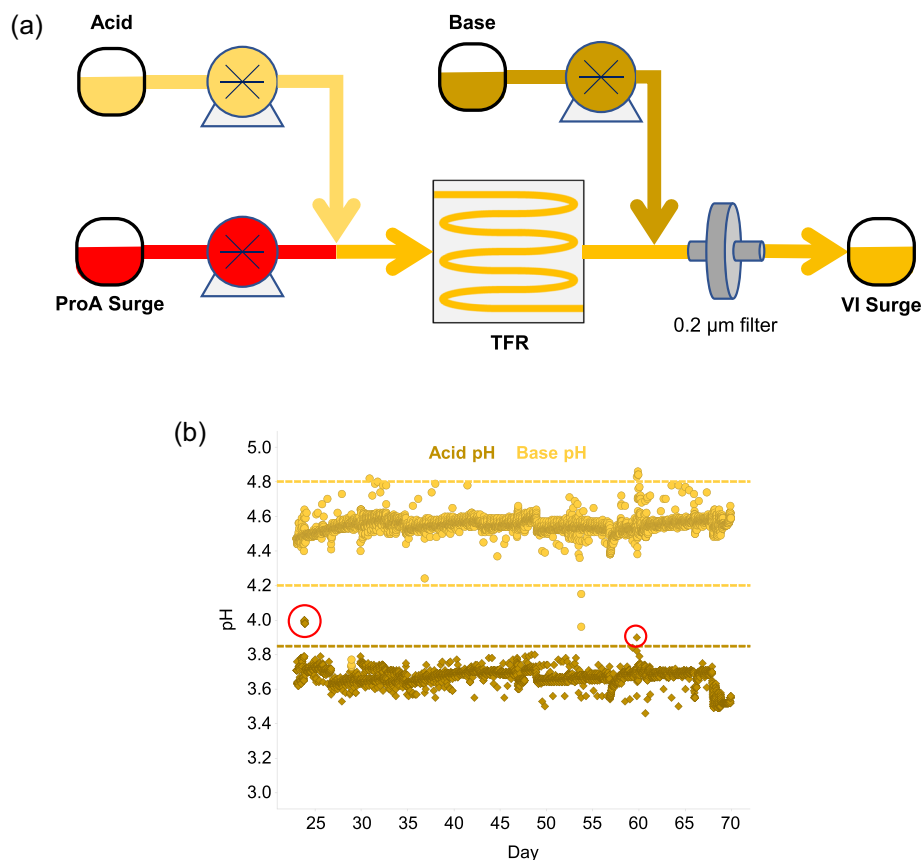


FIGURE 4 Summary of results for the VI operation. (a) Schematic of the in-line low pH VI operation. (b) pH trends for the VI operation. The pH post-acid addition is shown in the darker diamonds, while the post-base pH is shown in the lighter circles. The two time points where the acid pH went above the limit of 3.85 are shown in the red circles. ProA, Protein A; TFR, tubular flow reactor; VI, virus inactivation

points during the campaign: once on Day 23 as the system was still settling into steady-state and a second time on Day 60, which corresponds to the known process disturbance on ProA (Section 3.6.2). For this campaign, no action was taken when the inactivation pH went out-of-specification, but in the future, material could be diverted until the pH returned to acceptable range.

3.3.3 | Polishing chromatography

For this demonstration, a multi-column polishing chromatography train was selected to achieve the design goals of minimizing product residence time and minimizing surge vessels. A schematic representation of the polishing chromatography operation is shown in Figure 2d. The combined Capto Adhere/Capto Phenyl, polishing chromatography operation was integrated on Day 23 after the VI operation had reached a steady state. Over the course of the 47 days of operation, each of the individual columns was cycled ~80 times. The average cycle time was ~6.5 h, and ranged from around 5–8 h, depending on the ProA cycle time. The column loading varied from ~250–450 g/L, with an average of

~370 g/L. A snapshot of the chromatogram for the polishing chromatography can be found in Figure 2, along with an overlay of the column flowthrough chromatograms. For clarity, only a portion of the campaign (corresponding to the first 7 days of DS generation) is shown in the chromatogram, and every five cycles are included in the overlay.

For the most part, the column flowthrough profile was consistent for the duration of the run. The main difference was in the length of the flowthrough peak, which varied due to the differences in ProA cycle time. There were a small number of runs (~4%) with periods of decreased UV during the flowthrough, but no obvious trends were observed. Overall, the polishing column performance was within expectation for the entire campaign.

3.3.4 | Virus filtration

For this demonstration, the virus filtration was conducted in tangential flow mode, as illustrated in the schematic in Figure 6a. This mode of operation was selected as it can lead to greater filter throughput due to decreased filter fouling (Vetter et al., 2019). The applied pressure to the virus filtration loop was automated to control

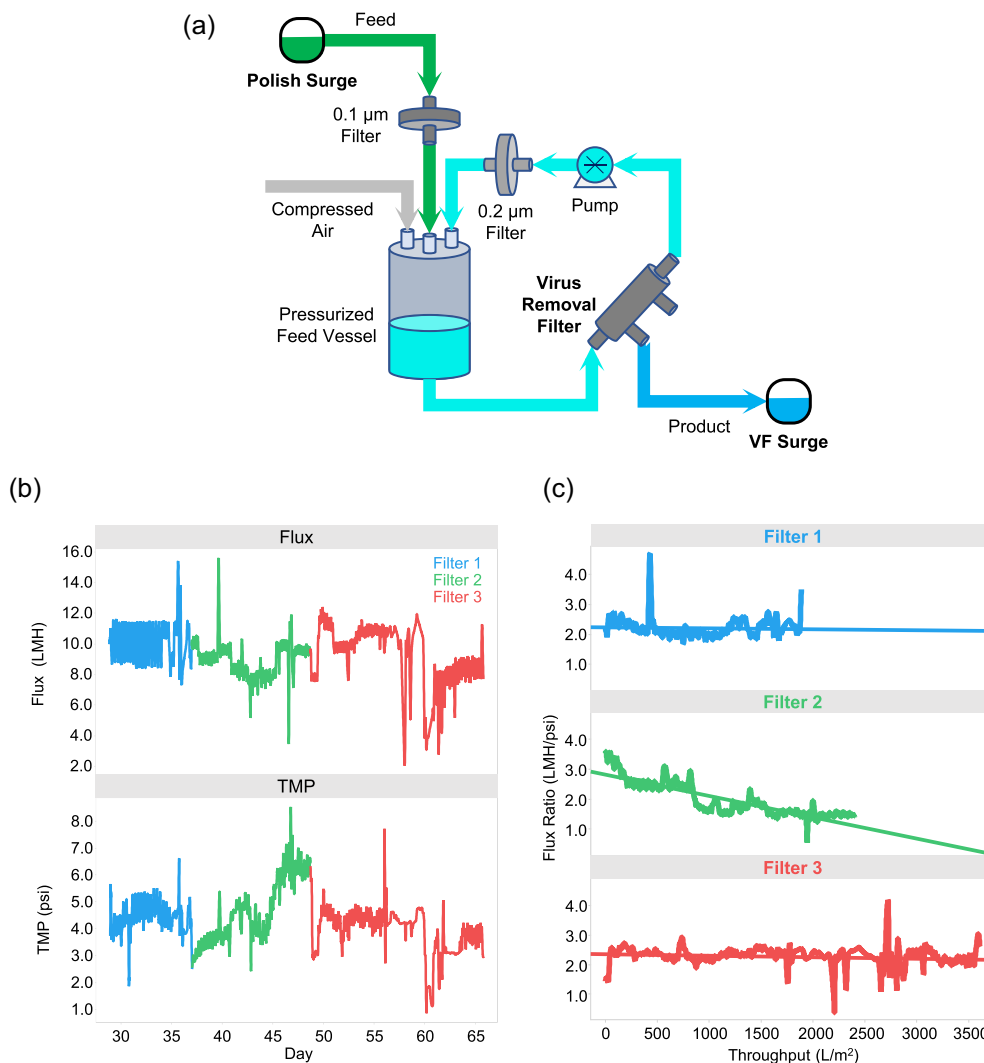


FIGURE 5 Summary of results for the virus filtration operation. (a) Schematic of the tangential flow virus removal filtration operation. (b) Flux and TMP trends over time for the virus filtration operation. The three virus filters are indicated in blue, green, and red, respectively. (c) Flux ratio versus volumetric throughput for the three virus filters. Filter 2 exhibited a decrease in filter performance, but Filters 1 and 3 exhibited no decrease in performance. TMP, transmembrane pressure; VF, virus filtration

the flux through the filter and match the incoming product flowrate, ensuring no accumulation of material in the VF loop.

The VF operation was integrated on Day 29 and executed for a total of 37 days. The duration of the VF operation was split between three separate VRFs (Filters 1, 2, and 3), as shown in Figure 6. The filter flux (in LMH) and transmembrane pressure (TMP; in psi) for the virus filtration operation is shown in Figure 6b. There was some variability in flux over time, primarily due to variability in the flow measurement. Overall, the average volumetric flux was approximately 9 LMH. For Filters 1 and 3, the TMP was relatively constant at ~4 psi. Filter 2 was also operating at ~4 psi for the first 7 days, but the pressure increased, progressively reaching ~6 psi and remaining steady for the remaining ~5 days.

However, since neither the flux nor the pressure was truly constant for any given filter, it is relatively difficult to interpret filter performance from these trends alone. Instead, the flux to a pressure ratio (in LMH/psi)

was calculated and plotted versus the total volumetric throughput (in L/m²) for each filter and is shown in Figure 6c. The flux to pressure ratio is a measure of the filter resistance to flow and is a general metric to track filter performance, with a decrease in the ratio suggestive of filter fouling. For Filters 1 and 3, the flux to pressure ratio was consistent throughout the filter lifetime, suggesting that there was minimal filter fouling. Filter 2 exhibited a persistent decrease in flux to pressure ratio throughout the approximately 2 weeks of operation. An investigation into the Filter 2 performance concluded that the fouling was likely due to a buildup of particulates in the recirculation loop due to fouling of the 0.2 μm filter in the loop. After correcting the issue with the 0.2 μm filter, the expected performance was re-established for Filter 3. The results for Filter 3 were particularly encouraging, with a sustained operation for >2 weeks, a volumetric throughput of >3500 L/m², and a mass throughput of >30 kg/m² with no indication of fouling. These very high mass and volumetric loadings showed no operational issues, but the virus

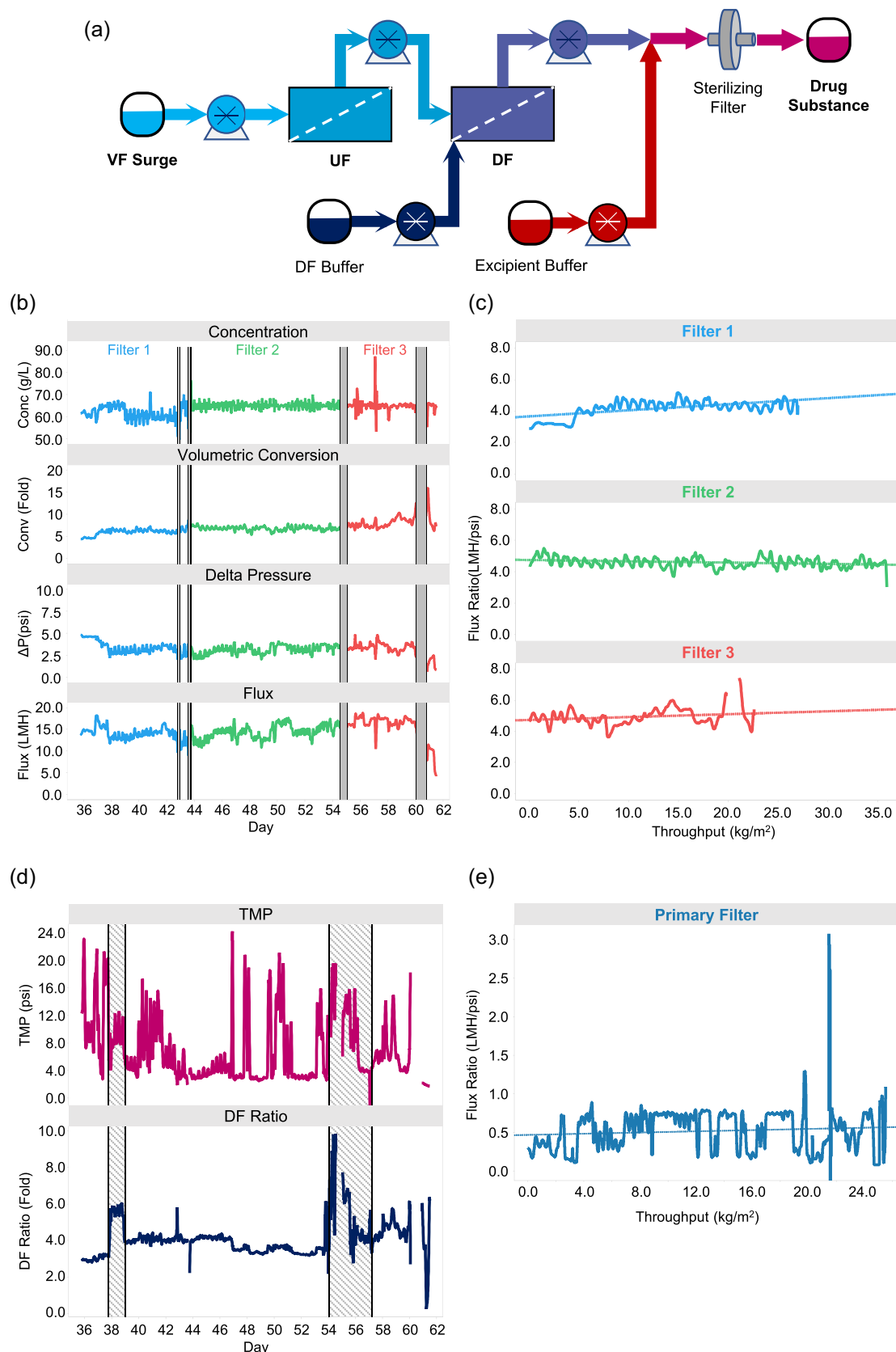


FIGURE 6 Summary of results for the UF and DF operations. (a) Schematic of the single-pass UF, single-pass DF, and in-line formulation operations. (b) Retentate concentration, volumetric conversion, delta pressure, and permeate flux trends over time for the UF operation. The three UF filters are indicated in blue, green, and red, respectively. Gray boxes indicate times corresponding to known process disturbances. (c) Flux ratio versus mass throughput for the three UF filters. No obvious indications of fouling were observed out to very high mass throughputs. (d) TMP and DF ratio trends over time for the DF operation. Gray-shaded regions indicated times when two DF filters were run in parallel. (e) Flux ratio versus mass throughput for the DF filter that operated for the duration of the run. No obvious indication of filter fouling was observed. DF, diafiltration; TMP, transmembrane pressure; UF, ultrafiltration; VF, virus filtration

removal capability of the system at these loadings would need to be confirmed with virus validation studies.

3.3.5 | Ultrafiltration, diafiltration, and formulation

For this demonstration, the ultrafiltration, diafiltration, and formulation unit operations were directly integrated with no intermediate surge vessels, as shown in the schematic in Figure 3a. The integrated operation consisted of a single-pass tangential flow filtration (TFF) filter (Pall In-line Concentrator) for the UF operation, a prototype single-pass diafiltration (SP-DF) filter (Sartocon SP, Sartorius) for the DF operation, and in-line addition of excipients for the formulation operation.

The UF and DF operations were initiated on Day 36 and executed for a total of 26 days. There were two process disturbances associated with the UF operation (at Days 43 and 55) that are described in detail in the process disturbances section (Section 3.6.1). The UF and DF operations were also impacted by the ProA process disturbance on Day 60. Data from the UF and DF operations during these process disturbances are not included in Figure 3 and indicated by gray boxes. Additionally, the disturbances on Days 43 and 55 necessitated unplanned filter swaps for the UF operation. Due to this, the 26 days of integrated operation were split among/across three separate ultrafilters, as indicated in Figure 3b.

Over the course of the run, the feed for the UF operation had a relatively consistent product concentration of around 8–10 g/L, while the inlet flowrate varied from approximately 15–25 ml/min to match the VF operation output (data not shown). The single-pass volumetric conversion, the UF retentate concentration (based on in-line measurement), delta pressure, and permeate flux are shown in Figure 3b. The UF retentate concentration was relatively consistent within the target range of 65 ± 5 g/L, indicating that the feedback control for concentration was working as expected. The volumetric conversion was relatively consistent from ~ 6 – $9\times$ over the course of the run. The delta pressure was quite consistent over the course of the run at ~ 3 – 5 psi. While the flux data is relatively noisy, there were also no obvious trends in the flux over time.

Finally, the flux to a pressure ratio (described previously in Section 3.3.4) versus mass throughput is plotted in Figure 6c. For all three filters, the flux to pressure ratio was consistent with no obvious downward trends, suggesting that there was no filter fouling occurring. Of note, approximately 35 kg/m² of material was processed over Filter 2 with no evidence of decreasing filter performance.

Overall, the performance of the ultrafiltration operation was consistent throughout the course of the run. In certain respects, the performance exceeded expectations. The high mass throughput (>35 kg/m²) with no evidence of fouling (even without intermediate caustic cleaning) is significantly higher than comparable batch processes, where a typical mass loading is around 1 kg/m². Additionally, the low delta pressures (~ 5 psi) suggest that the single-pass conversion could have been achieved with less total filter area. These two results suggest that continuous single-pass ultrafiltration could be achieved with much less filter area and much smaller equipment than a comparable batch ultrafiltration operation.

The prototype single-pass diafiltration cassette from Sartorius (Sartocon SP) is designed to achieve a comparable level of buffer exchange to a traditional diafiltration operation without the need for recirculation. Preliminary testing before the integrated campaign confirmed that the level of buffer exchange was comparable to traditional diafiltration (data not shown). This testing also determined that a single prototype cassette (0.2 m² filter area) would not be sufficient to achieve the target of seven diavolumes of exchange at the expected mass flow of the integrated campaign (details in the Supporting Information). To achieve the required buffer exchange, the DF filter area was increased by utilizing two cassettes in parallel.

The inlet pressure and diafiltration ratio (buffer flow/product flow) versus time are shown in Figure 3d. Maintaining consistent flow between the two filter cassettes proved to be a significant challenge, so the majority of the run was executed with a single filter at a lower target diafiltration ratio of 3–4 $\times$. The regions where two filters were run in parallel are highlighted in the figure. When both filters were included in-line, it was possible to achieve the target diafiltration ratio. Despite these difficulties, the diafiltration ratio was well controlled around the setpoint at any given time. The inlet pressure trace exhibited significant and unexplained spikes that do not seem to be correlated with any known process disruptions. No definitive root cause for the pressure spiking was identified.

The flux to pressure ratio versus mass throughput for the DF filter that was in-line for the duration of the campaign is shown in Panel E. The flux to pressure ratio for the DF filter is much noisier than the UF data, due largely to the noise in the pressure signal and the difficulties in measuring the lower flow rates for this operation. Despite the noise, there is no obvious downward trend in flux/pressure ratio up to 25 kg/m² mass loading, suggesting that there was minimal fouling in the filter.

Overall, the DF operation performance was within expectation for the duration of the campaign. The performance was fairly consistent but not as consistent as the UF operation. This could be due to the use of a preliminary prototype cassette, lack of experience with this operation, and also with difficulty in measurement and control of very low flow rates. Despite some of the challenges, the absence of fouling at high mass loadings for the DF filter is suggestive that the required filter area could be significantly reduced for the continuous DF operation relative to a comparable batch operation.

The formulation operation was initiated on Day 37 and continued for 25 days. There were no process performance indicators associated with the formulation, so discussion of the results of this operation are included in the DS generation and product quality section (Section 3.5). The UF, DF, and formulation operations were terminated on Day 62 due to a process disturbance described in the process disturbance section (Section 3.6.3).

3.4 | Overall process performance and intermediate quality

Selected shared performance characteristics for all the unit operations can be found in Figure 7. The first panel shows an estimate of the mass

flux (in grams per day) for the respective unit operations versus time. The mass flux was calculated using the in-line flow measurements and either in-line or off-line concentration measurements. The concentration between off-line sampling points was interpolated from the available data using a cubic spline fit. The mass flux data is relatively noisy, but in general, the trends are consistent with expectations. Focusing on the region where all units were integrated, it is possible to use the mass flux for each unit operation to estimate the recovery for each step and the overall process. The average mass flux and product

recoveries for each operation are shown in the figure. The average overall process recovery was ~70%, which is consistent with the expectations for the batch purification process for this molecule.

While this analysis is instructive, it does highlight the difficulties in calculating product recoveries for integrated and continuous processes. In particular, the recovery is less than expected for the polishing columns (~65% vs. 90%) and higher than expected for the VF operation (~120% vs. ~100%). In the future, additional in-line concentration sensors and more reliable flow measurements,

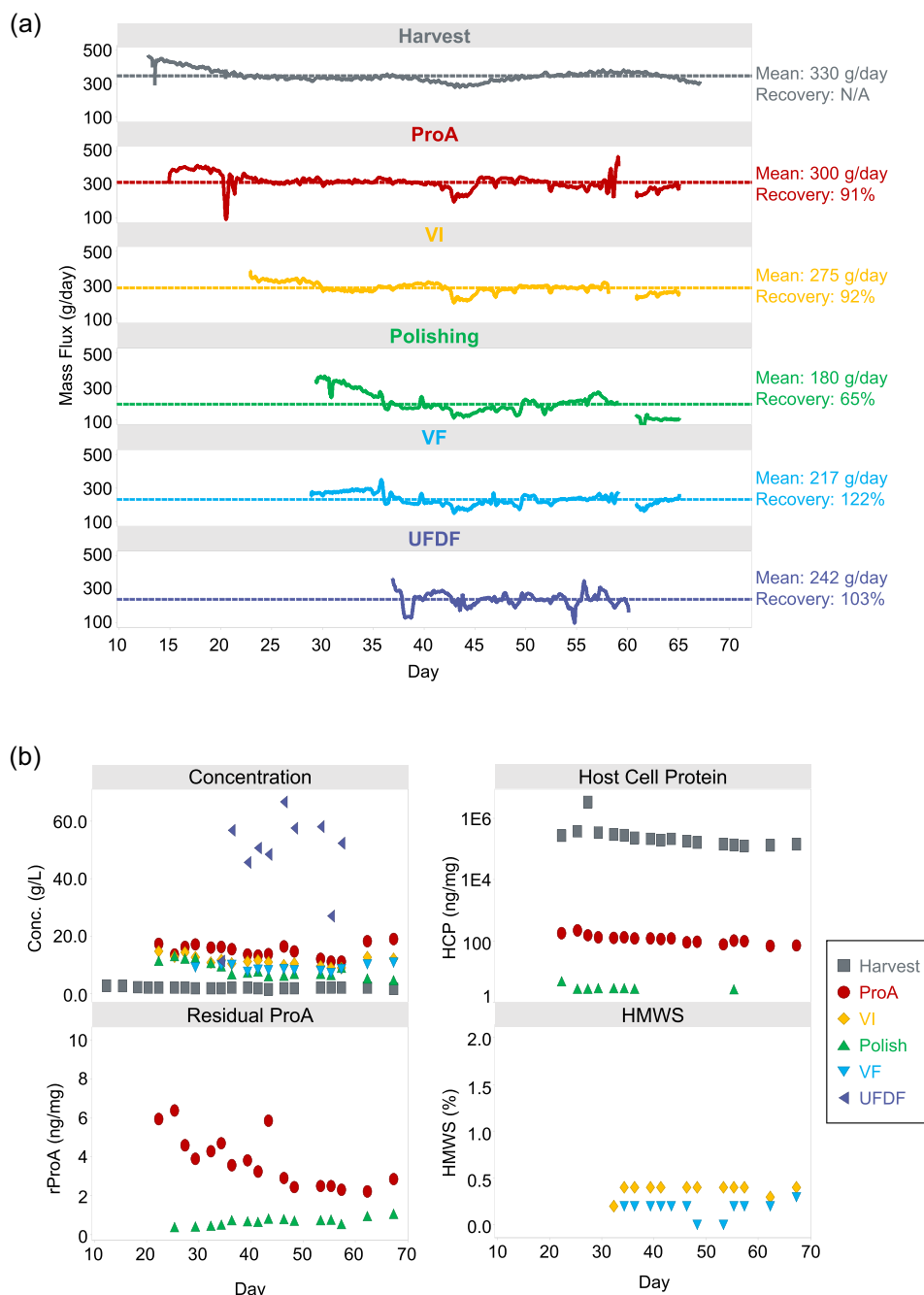


FIGURE 7 Summary of overall process performance and intermediate product quality. (a) Mass flux trends over time for each of the unit operations. The mean mass flux for the 25-day period of DS generation is shown in the horizontal dashed lines and to the right of the plot. The average operation recovery (yield) is also indicated. (b) Product concentration, host cell protein, residual ProA, and HMWS trends over time for the different unit operations. DF, diafiltration; DS, drug substance; HCP, host cell protein; HMWS, high molecular weight species; ProA, Protein A; rProA, residual Protein A; UF, ultrafiltration; VF, virus filtration; VI, virus inactivation

especially for highly concentrated streams, would make this analysis more robust.

Key product quality attributes for the process intermediates are shown in Figure 7b. The product concentration was relatively consistent over time for each of the intermediates. The one unexpected result was that the Phenyl flowthrough was a lower concentration than the VF permeate, which could be due to a sampling issue. The HCPs and rProA values were consistent and within expectation for this process. Finally, the HMWS values were low throughout the course of the run and for each intermediate tested. Overall, the purification process performed consistently with the expectations for the batch process for this molecule. A table with the detailed data from Figure 7 is included in the Supporting Information.

3.5 | DS generation and product quality

The DS generation was initiated on Day 37 by processing the formulated bulk through a sterilizing grade filter. The DS generation was carried out for a total of 25 days and split into six separate batches, with DS switching occurring when each collection vessel was full. Each of the six batches was submitted for a panel of product

quality tests based on the release testing for this molecule. The results of the DS product quality tests are shown in Figure 8.

The charge isoform distribution, purity, HCPs, rProA, HMWS, and pH were consistent for each of the six batches and met the release specifications for this molecule. The product concentration and osmolality both varied outside of the release specifications for some of the batches. The variability in the osmolality was likely caused by variability in the formulation operation due to the difficulty in maintaining consistent flow rate ratios at the low flow rates of this step. Variability in flow ratio would lead to variability in sugar concentration in the final product, which was the primary driver of the osmolality. Variability in the formulation operation partially, but not fully, explains the variability in the final product concentration. It is likely that the product concentration coming into the formulation operation was also more variable than expected, potentially due to dilution/concentration during the diafiltration step. A more robust control strategy for this operation will need to be developed in the future to ensure that the concentration specifications are met.

Over the course of the 25 days, a total of 4.9 kg of DS was generated, with an average productivity of around 200 g of mAb per day (2 g DS/L BRX/day). Overall, the DS product quality was within expectation and acceptable, except for the attributes mentioned previously.

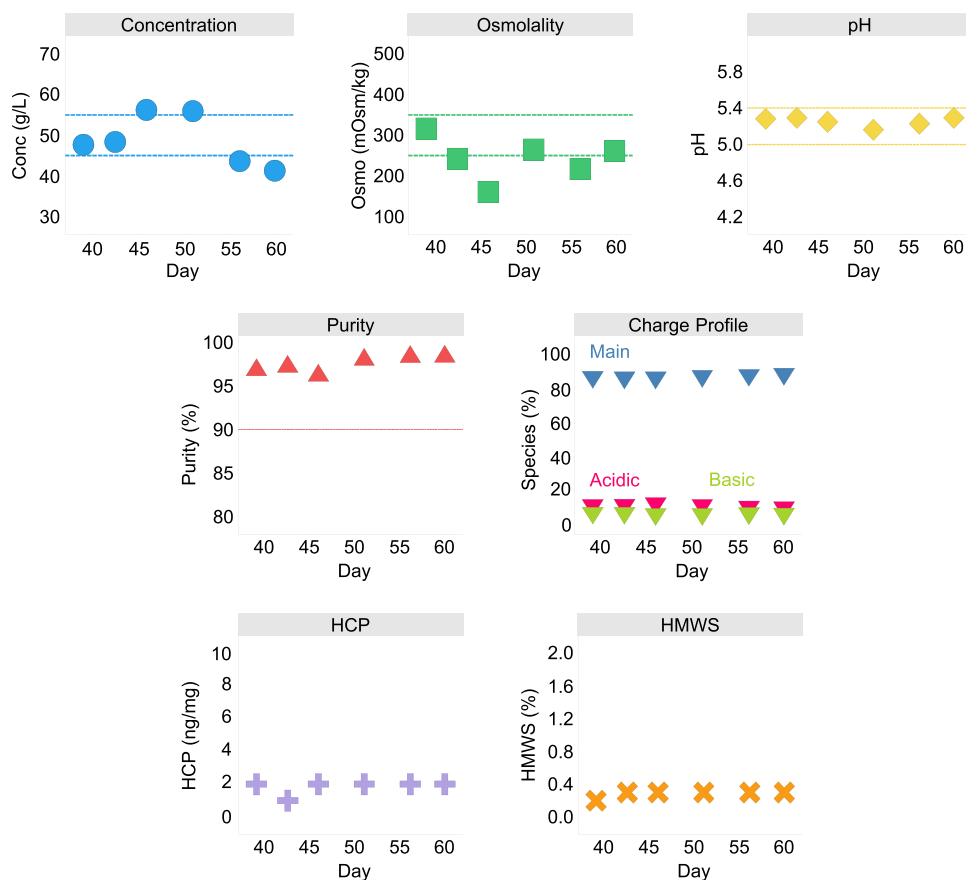


FIGURE 8 Product quality summary for the six DS batches. Values for individual batches are shown as points, with acceptance levels indicated by dashed lines. The pH, purity, charge profile, HCP, and HMWS met acceptance levels for all batches. The product concentration and osmolality did not meet the specification for several batches. DS, drug substance; HCP, host cell protein; HMWS, high molecular weight species

3.6 | Known process disturbances

A summary of known process disturbances and interruptions is presented in this section.

3.6.1 | Days 43 and 55-UF process disturbance

On both Days 43 and 55, the UF/DF waste vessel (connected to the permeate lines) was manually drained by the operator. Failure to turn off the pump when the vessel was emptied caused a partial vacuum to be pulled on the permeate line for both filters. This caused air to be drawn into the UF filter, which could not be subsequently purged from the system, requiring a UF filter switch. This disturbance impacted the UF and DF operations.

3.6.2 | Day 60-ProA disturbance

On approximately Day 60 of the campaign, a fresh batch of ProA elution buffer was prepped using the in-line conditioning skid. The pH of this buffer was above the specification limit, potentially due to drift in the ILC skid pH sensor over time. The higher elution pH caused the ProA elution peak to drag and increased the elution volume. This increased elution volume eventually caused the VI surge vessel to overflow, leading to a temporary pause in the VI operation and subsequent downstream operations. On approximately Day 62, the Protein A method was modified to cut the elution peak earlier and prevent the downstream effects. This disturbance impacted the VI, polishing, VF, UF, and DF operations.

3.6.3 | Day 62—UF, DF, formulation termination

On Day 62, the process closure in the UF retentate line was breached due to a loss of integrity (leak) in the FlowVPE. This was likely caused by wear in the O-rings in the FlowVPE. At that time, it was determined that replacing the parts and re-establishing process closure was not feasible, so the impacted operations (UF, DF, formulation) and DS generation were terminated. The remaining unit operations were continued to generate additional data and process experience.

3.7 | Process intensification

The process described above was used to generate approximately 5 kg of the DS over the course of 25 days, with an average productivity of 200 g/day (2 g/L BRX/day) of the DS. Extrapolating this productivity, a fully utilized manufacturing plant implementing this process (100-L perfusion bioreactor) would be expected to produce approximately 100 kg of DS per year. A roughly equivalent amount of DS could be generated from a fully utilized 1000-L traditional fed-batch facility, assuming a bioreactor titer of 5 g/L. A comparison of

the equipment sizing for the fed-batch process and the process described here can be found in Table 2 below.

As shown in Table 2, there is a significant reduction in the size of both the equipment and the vessels for the ICB process, with the reduction varying from ~2× smaller for the VRF to 100× smaller for the process intermediate vessels. The overall vessel volume is decreased approximately 20-fold for the ICB process relative to the fed-batch process. This analysis does not consider the buffer tank volume, which is dependent on the buffer generation/delivery strategy and can vary for both types of processing, or the scale of supporting equipment (e.g., infrastructure for utilities, pumps, HVAC) associated with each process. However, even from this analysis, it is clear that the ICB process can generate an equivalent amount of DS in a substantially smaller footprint.

4 | CONCLUSIONS

We have presented here a successful proof-of-concept for an integrated, end-to-end continuous pilot-scale DS manufacturing process for a mAb. This process integrated all the unit operations typically found in an mAb production process, including the final viral filtration, UF/DF, and formulation operations. This proof-of-concept process was designed to incorporate: a high degree of automation to minimize operator interactions; a closed system (where feasible) to enable operation in a multiproduct ballroom space; a continuous flow of product from the bioreactor to DS with minimal hold-up time in the process; and pilot-scale operation to identify potential failure modes in advance of GMP implementation. The entire continuous system, from the bioreactor to formulated DS, was integrated for a total of 25 days and was used to successfully generate a total of 4.9 kg of the DS.

TABLE 2 Comparison of unit operation scales for the ICB process in this demonstration and a comparable fed-batch process

Equipment	Fed-batch process	ICB process	Fold reduction
Bioreactor	1000 L	100 L	10×
Harvest vessel	1000 L	100 L	10×
Capture column	25 L	2 × 1 L	12.5×
Intermediate vessels (×4)	500 L	5 L	100×
VI vessel	500 L	5 L	100×
Polishing columns	2 × 5 L	4 × 0.2 L	12.5×
Virus removal filter	1 m ²	0.5 m ²	2×
UF/DF membrane	5 m ²	0.5 m ²	10×
Total vessel volume	4500	220	~20×

Note: Both processes generate ~100 kg DS per year. Significant reduction in the equipment sizing can be achieved in the ICB process.

Abbreviations: DF, diafiltration; DS, drug substance; ICB, integrated and continuous bioprocessing; UF, ultrafiltration; VI, virus inactivation.

Integration from bioreactor through continuous viral filtration was achieved for a total of 37 days. The product quality was consistent over the duration of the run and met the release specifications for the test molecule, except for the product concentration and osmolality, which varied due to inconsistencies in the formulation operation.

While this proof-of-concept was successful, several opportunities for future improvement were identified. The most important opportunity would be improvements in the control of the final formulation operation. Because of the low flow rates in the final product stream, it is difficult to rely on currently available flow sensors and pumps to ensure the correct addition of excipients. Another future improvement would be to apply Process Analytical Technology more broadly throughout the process, both for process monitoring and to improve the robustness of control. A final opportunity for improvement would be to move to a 100% single-use system based on pre-sterilized (not sanitized) process equipment, allowing for the process to be truly closed instead of functionally closed.

In addition to opportunities for improvement, several gaps would likely need to be addressed before this system could be used in commercial manufacturing. These gaps include the following: availability of closed purification equipment at smaller scales (particularly chromatography columns); reliability of flowmeters at very low flow rates; incorporation of control elements across multiple integrated unit operations; and strategies for handling out-of-specification material. Despite these remaining gaps, this study has demonstrated that it is possible to achieve an end-to-end integrated and continuous mAb production process at a pilot-scale with currently available process technology. To the authors' knowledge, this is the first such demonstration of this type reported in the literature. This marks an important achievement on the path toward commercializing ICB processes and realizing the significant benefits that are enabled by that transition.

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AUTHOR CONTRIBUTIONS

Michael J. Coolbaugh, Chad T. Varner, Tarl A. Vetter, Emily K. Davenport, Brad Bouchard, Marcus Fiadeiro, Jason Walther, Rohan Patil, and Kevin Brower conceived and planned the study. Michael J. Coolbaugh, Chad T. Varner, Tarl A. Vetter, Emily K. Davenport, and Brad Bouchard executed the experiments. Michael J. Coolbaugh, Chad T. Varner, Tarl A. Vetter, Emily K. Davenport, Brad Bouchard, Marcus Fiadeiro, Nihal Tugcu, Jason Walther, Rohan Patil, and Kevin Brower contributed to the interpretation of the results. Michael J.

Coolbaugh wrote the manuscript with support from Chad T. Varner, Tarl A. Vetter, Emily K. Davenport, Rohan Patil, Jason Walther, and Kevin Brower. Kevin Brower supervised the project. All authors reviewed the results and approved the final version of the manuscript.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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SUPPORTING INFORMATION

Additional Supporting Information may be found online in the supporting information tab for this article.

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